

minimality of $\text{ter}(\mathcal{P})$. Hence $\text{ter}(\mathcal{P}'') \subseteq \{v_3, \dots, v_m\}$. But now $\mathcal{P}''$ and the trivial path $\{v_1\}$ together form a path cover of G that contradicts the minimality of $\text{ter}(\mathcal{P})$. $\square$

As a corollary to Theorem 2.5.1 we obtain a classical result from the theory of partial orders. Recall that a subset of a partially ordered set $(P, \leq)$ is a *chain* in P if its elements are pairwise comparable; it is an *antichain* if they are pairwise incomparable.

chain
antichain

Corollary 2.5.2. (Dilworth 1950)

In every finite partially ordered set $(P, \leq)$, the minimum number of chains with union P is equal to the maximum cardinality of an antichain in P .

Proof. If A is an antichain in P of maximum cardinality, then clearly P cannot be covered by fewer than $|A|$ chains. The fact that $|A|$ chains will suffice follows from Theorem 2.5.1 applied to the directed graph on P with the edge set $\{(x, y) \mid x < y\}$. $\square$

Exercises

- Let M be a matching in a bipartite graph G . Show that if M is sub-optimal, i.e. contains fewer edges than some other matching in G , then G contains an augmenting path with respect to M . Does this fact generalize to matchings in non-bipartite graphs?
- (continued)
Describe an algorithm that finds, as efficiently as possible, a matching of maximum cardinality in any bipartite graph.
- Show that if there exist injective functions $A \rightarrow B$ and $B \rightarrow A$ between two infinite sets A and B then there exists a bijection $A \rightarrow B$.
- Moving alternately, two players jointly construct a path in some fixed graph G . If $v_1 \dots v_n$ is the path constructed so far, the player to move next has to find a vertex v_{n+1} such that $v_1 \dots v_{n+1}$ is again a path. Whichever player cannot move loses. For which graphs G does the first player have a winning strategy, for which the second?
- Derive the marriage theorem from König's theorem.
- Let G and H be defined as for the third proof of Hall's theorem. Show that $d_H(b) \leq 1$ for every $b \in B$, and deduce the marriage theorem.
- Does our first proof of the marriage theorem use the assumption that the graph is finite? If so, can it be adapted so that it works for infinite graphs too?

8. Let k be an integer. Show that any two partitions of a finite set into k -sets admit a common choice of representatives.
9. Let A be a finite set with subsets $A_1, \dots, A_n$, and let $d_1, \dots, d_n \in \mathbb{N}$. Show that there are disjoint subsets $D_k \subseteq A_k$, with $|D_k| = d_k$ for all $k \leq n$, if and only if

$$\left| \bigcup_{i \in I} A_i \right| \geq \sum_{i \in I} d_i$$

for all $I \subseteq \{1, \dots, n\}$.

- 10.⁺ Prove that in an n -set X there are never more than $\binom{n}{\lfloor n/2 \rfloor}$ subsets such that none of these contains another.
(Hint. Construct $\binom{n}{\lfloor n/2 \rfloor}$ chains covering the power set lattice of X .)
11. Let G be a bipartite graph with bipartition $\{A, B\}$. Assume that $\delta(G) \geq 1$, and that $d(a) \geq d(b)$ for every edge ab with $a \in A$. Show that G contains a matching of A .
12. Find a bipartite graph with a set of preferences such that no matching of maximum size is stable and no stable matching has maximum size. Find a non-bipartite graph with a set of preferences that has no stable matching.
- 13.⁻ Consider the algorithm described in the proof of the stable marriage theorem. Observe that once a vertex of B is matched, she remains matched and gets happier with every change of her matching edge. On the other hand, show that the sequence of matching edges incident with a given vertex of A makes this vertex unhappier with every change (disregarding the interim periods when he is unmatched).
14. Show that all stable matchings of a given graph cover the same vertices. (In particular, they have the same size.)
- 15.⁺ Show that the algorithm in our proof of Theorem 2.1.4 produces a matching M such that no other stable matching makes any vertex in A happier or any vertex in B unhappier than he or she is in M . Consider only matched vertices for happiness.
- 16.⁺ Show that the following ‘obvious’ algorithm need not produce a stable matching in a bipartite graph. Start with any matching. If the current matching is not maximal, add an edge. If it is maximal but not stable, insert an edge that creates instability, deleting any current matching edges at its ends.
17. Show that the union of two partial orderings $\leq_1, \leq_2$ of a finite set P has a ‘dominating antichain’, a set $A \subseteq P$ such that no two elements of A are related in either $\leq_1$ or $\leq_2$ and for every $x \in P$ there exists an $a \in A$ such that $x \leq_1 a$ or $x \leq_2 a$. Deduce Theorem 2.1.4.
18. Find a set S for Theorem 2.2.3 when G is a forest.

19. A graph G is called (vertex-) *transitive* if, for any two vertices $v, w \in G$, there is an automorphism of G mapping v to w . Using the observations following the proof of Theorem 2.2.3, show that every transitive connected graph of even order contains a 1-factor.
- 20.⁺ Show that a graph G contains k independent edges if and only if $q(G - S) \leq |S| + |G| - 2k$ for all sets $S \subseteq V(G)$.
- 21.⁻ Find a cubic graph without a 1-factor.
- 22.⁺ Derive the marriage theorem from Tutte's theorem.
- 23.⁻ Disprove the analogue of König's theorem (2.1.1) for non-bipartite graphs, but show that $\mathcal{H} = \{K^2\}$ has the Erdős-Pósa property.
24. Let T be a tree and $\mathcal{T}$ a set of subtrees of T . Show that the maximum number of disjoint trees in $\mathcal{T}$ equals the least cardinality of a set X of vertices such that $T - X$ contains no tree from $\mathcal{T}$.
25. For cubic graphs, Lemma 2.3.1 is considerably stronger than the Erdős-Pósa theorem. Extend the lemma to arbitrary multigraphs of minimum degree ≥ 3 , by finding a function $g: \mathbb{N} \rightarrow \mathbb{N}$ such that every multigraph of minimum degree ≥ 3 and order at least $g(k)$ contains k disjoint cycles, for all $k \in \mathbb{N}$. Alternatively, show that no such function g exists.
26. Given a graph G , let $\alpha(G)$ denote the largest size of a set of independent vertices in G . Prove that the vertices of G can be covered by at most $\alpha(G)$ disjoint subgraphs each isomorphic to a cycle or a K^2 or K^1 .
27. Show that if G has two edge-disjoint spanning trees, it has a connected spanning subgraph all whose degrees are even.
28. In the proofs of Theorems 2.4.1, 2.4.3 and 2.4.4, there is exactly one place where we use that we are working with multigraphs. Where is it?
29. Find the error in the following short 'proof' of Theorem 2.4.1. Call a partition *non-trivial* if it has at least two classes and at least one of the classes has more than one element. We show by induction on $|V| + |E|$ that $G = (V, E)$ has k edge-disjoint spanning trees if every non-trivial partition of V into r sets (say) has at least $k(r - 1)$ cross-edges. The induction starts trivially with $G = K^1$ if we allow k copies of K^1 as a family of k edge-disjoint spanning trees of K^1 . We now consider the induction step. If every non-trivial partition of V into r sets (say) has more than $k(r - 1)$ cross-edges, we delete any edge of G and are done by induction. So V has a non-trivial partition $\{V_1, \dots, V_r\}$ with exactly $k(r - 1)$ cross-edges. Assume that $|V_1| \geq 2$. If $G' := G[V_1]$ has k disjoint spanning trees, we may combine these with k disjoint spanning trees that exist in G/V_1 by induction. We may thus assume that G' has no k disjoint spanning trees. Then by induction it has a non-trivial vertex partition $\{V'_1, \dots, V'_s\}$ with fewer than $k(s - 1)$ cross-edges. Then $\{V'_1, \dots, V'_s, V_2, \dots, V_r\}$ is a non-trivial vertex partition of G into $r + s - 1$ sets with fewer than $k(r - 1) + k(s - 1) = k((r + s - 1) - 1)$ cross-edges, a contradiction.

30. A graph G is called *balanced* if $\varepsilon(H) \leq \varepsilon(G)$ for every subgraph $H \subseteq G$.
 - (i) Find a few natural classes of balanced graphs.
 - (ii) Show that the arboricity of a balanced graph is bounded above by its average degree. Is it even bounded by ε ? Or by $\varepsilon + 1$?
 - (iii) Characterize, in terms of the balanced graphs or otherwise, the graphs G such that $\varepsilon(H) \geq \varepsilon(G)$ for every induced subgraph $H \subseteq G$.
31. Rephrase König's and Dilworth's theorems as pure existence statements without any inequalities.
32. Prove the undirected version of the theorem of Gallai & Milgram (without using the directed version).
33. Derive the marriage theorem from the theorem of Gallai & Milgram.
34. Show that a partially ordered set of at least $rs + 1$ elements contains either a chain of size $r + 1$ or an antichain of size $s + 1$.
35. Prove the following dual version of Dilworth's theorem: in every finite partially ordered set $(P, \leq)$, the minimum number of antichains with union P is equal to the maximum cardinality of a chain in P .
36. Derive König's theorem from Dilworth's theorem.
37. Find a partially ordered set that has no infinite antichain but is not a union of finitely many chains.

Notes

There is a very readable and comprehensive monograph about matching in finite graphs: L. Lovász & M.D. Plummer, *Matching Theory*, Annals of Discrete Math. **29**, North Holland 1986. Two other very comprehensive sources are A. Schrijver, *Combinatorial optimization*, Springer 2003, and A. Frank, *Connections in combinatorial optimization*, Oxford University Press 2011. All the references for the results in this chapter can be found in these books.

All the main theorems in this chapter are of a particularly attractive type, which they share with several other fundamental theorems in graph theory. They each assert that in some obvious inequality we can attain equality, or that some obvious potential obstruction to some desirable property must necessarily occur in all graphs that fail to have this property. Phrased in yet another way, these theorems provide easily checked 'certificates' for the absence of some desirable property, which comes with its own obvious certificate when it does hold. The theorems of König, Hall, Tutte, Dilworth, and Nash-Williams proved in this chapter can all be viewed in this way.

As we shall see in Chapter 3, König's Theorem of 1931 is no more than the bipartite case of a more general theorem that is also of this type, which is generally attributed to Menger (1927). However, Menger missed the bipartite case in his original proof. When Menger showed König his theorem and proof during a visit to Budapest in 1930, they seem to have noticed this gap. König